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Curriculum Vitae

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専門分野

群制御，人工知能，自己組織化，複雑システム

学歴

- テキサス工科大学 Ph.D. (Mathematics), 2014年
- 京都大学 修士（情報学），2009年
- 京都大学 学士（工学），2007年

職歴

- 広島大学 教授，2024年4月 –
- 大阪大学 招へい教授，2024年4月 – 2025年3月
- 大阪大学 准教授，2019年11月 – 2024年3月
- 奈良先端科学技術大学院大学 客員准教授，2020年4月 – 2021年3月
- 奈良先端科学技術大学院大学 助教，2017年3月 – 2019年10月
- ペンシルバニア大学 電気システム工学科 博士研究員，2014年10月 – 2017年2月

短期滞在

- Department of Mechanical Engineering, University of Hong Kong. 2018年, 2019年
- ICTEAM Institute, Université catholique de Louvain, Belgium. 2013年

受賞

- 2023年システム数理と応用研究会優秀論文賞 (2024)
- 計測自動制御学会 著述賞 (2021)
- 電子情報通信学会ネットワークシステム研究専門委員会第8回英語セッション奨励賞（2020）
- 電子情報通信学会情報ネットワーク研究会第26回情報ネットワーク研究賞（2019）
- Runner-up of the 2019 Best Paper Award, *IEEE Transactions on Network Science and Engineering* (2019)
- 計測自動制御学会 関西支部 支部長賞 奨励賞 (2018)
- 計測自動制御学会 制御部門 制御部門大会賞 (2018)
- Summer Dissertation/Thesis Research Award, Texas Tech University (2014)
- Cash Family Endowed Fellowship, Texas Tech University (2013)
- 計測自動制御学会 論文賞 (2012)

指導学生や共著者などの受賞

- 李, 計測自動制御学会優秀学生賞 (2023, 指導学生)
- 李, 大阪大学大学院情報科学研究科情報科学研究科賞 (2023, 指導学生)
- 藤岡, Finalist of SICE Annual Conference 2022 Young Author's Award (2022, 指導学生)
- 阿部, 第36回電気通信普及財団賞テレコムシステム技術学生賞 (2021, 共著者)
- 森, 2019年11月高信頼制御通信研究会ベストポスター賞 (2019, 指導学生)
- W. Mei, China Scholarship Council Scholarship for Doctoral Studies (2019, 指導学生)
- 蓼沼, Young Author Award Finalist in 5th IFAC Conference on Analysis and Control of Chaotic Systems (2018, 共著者)
- W. Mei, The IET Outstanding Reviewer Awards 2018 (2019, 指導学生)

研究助成（科学研究費など）

- 科学研究費基盤研究（A）「拡張排他待ち行列モデルを用いた渋滞・混雑の制御および行動変容方法の開発」（研究分担者，2026/04-）
- 科学技術振興機構戦略的創造研究推進事業 CRONOS 「視覚情報の時空間スパース性を活用した計測・通信・制御の協調設計」（研究参加者，2025/10-）
- 科学技術振興機構戦略的創造研究推進事業 CRONOS 「デジタルサイバネティクスのための次世代通信アーキテクチャの創出」（研究参加者，2025/10-）
- 科学研究費基盤研究（B）「費用最適な観測と介入を計画するためのトビバッタ対策立案プラットフォームの構築」（研究代表者，2025/04-）
- 科学研究費基盤研究（C）「目標指令値の損失に対処する切替フィードフォワード学習制御と移動体群の形成」（研究分担者，2025/04-）
- 科学研究費基盤研究（A）「選択的漁獲のための魚群行動の理解と制御」（研究分担者，2024/04-）
- 科学技術振興機構ムーンショット型研究開発事業ムーンショット目標8 海上豪雨生成で実現する集中豪雨被害から解放される未来「データ駆動手法を用いた制御手法の開発」（課題推進者，2023/12-）
- 科学技術振興機構ムーンショット型研究開発事業ムーンショット目標3 人・AIロボット・生物サイボーグの共進化による新ひらめきの世界「人・AIロボット・生物サイボーグの共進化による新ひらめきの世界」（研究参加者，2023/01-2024/09）
- 科学技術振興機構ムーンショット型研究開発事業ムーンショット目標8 気象制御のための制御容易性・被害低減効果の定量化「気象制御容易性の定量化」（課題推進者，2022/06-2023/11）
- 科学研究費基盤研究（A）「深層展開に基づく信号処理アルゴリズム構築論の深化と展開」（研究分担者，2022/04-2025/03）
- 科学研究費基盤研究（B）「異種マルチエージェントシステム制御における拡散的外部刺激の理論体系の確立と実検証」（研究代表者，2021/04-2025/03）
- 科学研究費基盤研究（B）「計測・通信品質が保証されない環境下の多目的フィードフォワード最適制御と強化学習」（研究分担者，2021/04-2024/03）
- 科学技術振興機構戦略的創造研究推進事業 CREST 「CyPhAI: Formal Analysis and Design of AI-intensive Cyber-Physical Systems」（研究参加者，2020/11-2026/03）
- 科学研究費若手研究「ネットワークにおける伝播の解析と制御：モチーフを活用した多項式時間アルゴリズム」（研究代表者，2018/04-2021/03）
- 科学研究費基盤研究（B）「計測や通信の品質が保証されない環境下での事象トリガ調整型2自由度制御系」（研究分担者，2018/04-2021/03）

研究助成（企業・財団など）

- 公益財団法人電気通信普及財団研究調査助成「グループ単位の情報共有を活かしたスワームロボット制御基盤の構築」（研究代表者，2025/04-2026/03）
- 2025年度国立情報学研究所公募型共同研究（研究企画会合公募型）「群制御の手法探索と社会実装へ向けた学際的な検討」（研究代表者，2025/04-2026/03）
- 2024年度国立情報学研究所公募型共同研究（研究企画会合公募型）「群制御の問題定式化，理論開拓，手法探索のための学際的な検討」（研究代表者，2024/04-2025/03）
- 令和5年度東北大学電気通信研究所共同プロジェクト研究 (区分B 萌芽的研究支援型)「牧羊犬のヒツジ追い現象におけるヒツジ・牧羊犬間の「駆け引き」メカニズムの解明」（共同研究者，2023/04-2024/03）
- 大阪大学データビリティフロンティア機構学際共創プロジェクト「機械学習と数理モデルによる細胞動態の解析手法の開発」（研究代表者，2022/10-2023/03）
- 一般財団法人テレコム先端技術研究支援センターSCAT研究費助成「多数端末収容のための無線LANに適した非直交多元接続向け自己組織型無線アクセス制御」（研究分担者，2022/04-2025/03）
- 大阪大学大学院情報科学研究科情報科学研究科スタートアッププログラム「深層展開によるモデルベース制御系設計：汎用性・有効性・使用性への挑戦」（研究代表者，2021/05-2022/03）
- 産学連携（2021）
- 公益財団法人電気通信普及財団研究調査助成「深層学習を用いた超高精度な行列因子分解」（研究調査代表者，2020/04-2021/03）
- モビリティ基盤数理研究ユニット動的ネットワーク制御チーム（メンバー，2019-）
- 公益財団法人電気通信普及財団海外渡航旅費援助（2019）

- 奈良先端科学技術大学院大学次世代融合領域研究推進プロジェクト「計測・解析・制御が融合したデータ駆動型細胞制御システムの開発」（共同研究者，2019/04-2021/03）
- 国立情報学研究所自由提案公募型共同研究「製品開発プロセスをロバスト化するための経営資源割当手法」（研究代表者，2019/04-2020/03）
- 国立情報学研究所自由提案公募型共同研究「幾何計画による非負システムに対する最適制御系設計」（研究代表者，2018/04-2019/03）
- 産学連携　株式会社ダイヘン（共同研究者，2017-2020）

研究業績

著書

[2] 角田祐輔, 今林亘, and 小蔵正輝, *Python*で始めるスワーム制御プログラミング．コロナ社, 2026.

[1] 永原正章, 岡野訓尚, 小蔵正輝, and 若生将史, ネットワーク化制御．コロナ社, 2019. **計測自動制御学会著述賞**.

著書（**book chapter**）

[4] K. Sakurama, K. Kashima, T. Ikeda, N. Hayashi, K. Hoshino, M. Ogura, and C. Zhao, "System-Control-Based Approach to Car-Sharing Systems," in *Advanced Mathematical Science for Mobility Society*. Springer Singapore, 2024, pp. 127–171.

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査読付き論文

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教育経験

広島大学大学院先進理工学研究科
- 複雑システム科学特論（2026–）

広島大学情報科学部
- 人工知能概論（2024–）
- 強化学習（2025–）
- 人工知能と機械学習（2024）

広島大学工学部
- 線形代数学II（2024–）

大阪大学大学院情報科学研究科
- バイオネットワーク基礎理論（2021–2023）
- バイオ情報工学入門（2020–2023）
- バイオネットワーク工学 (2020–2023)

大阪大学基礎工学部
- 情報数学基礎（2021–2023）
- 基礎工学のための情報学 I (2020–2023)
- 基礎工学 PBL(情報工学)（2020–2023）
- 情報科学ゼミナール（2020–2023）

奈良先端科学技術大学院大学
- 機械学習と知能制御（2019）
- 情報科学特別講義 (2018)
- 知能システム制御特論 (2017)

ペンシルバニア大学
- Convex Optimization in Systems and Control (Co-lecturer, Fall 2015)

テキサス工科大学
- 講師
  - Calculus II (Summer 2014, Spring 2014, Spring 2013)
  - Calculus I (Summer 2013, Fall 2012)
  - Trigonometry (Fall 2011)
  - College Algebra (Fall 2013, Spring 2012)
- ティーチングアシスタント
  - Advanced Calculus (Summer 2012)
  - Linear Algebra (Summer 2012)
  - Higher Mathematics for Engineers and Scientists I (Summer 2011)

京都コンピュータ学院
- 制御工学 (2009, 2009)
- 電気回路 (2008)
- データ構造 (2008)
- 数値解析 (2009, 2010)

京都大学（ティーチング・アシスタント）
- 自然現象と数学 (2009)
- 現代制御論 (2008, 2009)

学会活動，委員歴

- 電子情報通信学会 高信頼制御通信研究会，副委員長（2026, 2027）
- Program Committee Member, Sixth International Symposium on Complex Systems
- Associate Editor, 2026 SICE International Symposium on Control Systems
- Organized & Invited Sessions Chair, SICE Festival 2026 with Annual Conference
- Program Committee Member, 14th International Conference on Complex Networks and their Applications
- 計測自動制御学会 制御部門 制御理論部会，委員（2025, 2026）
- Associate Editor, The Journal of Engineering (2025–present)
- Technical Comittee Member, Technical Comittee on Fully Actuated System Theory and Applications (2025–present)
- 計測自動制御学会 制御部門 知的システム・ネットワーク統合のための適応学習制御調査研究会，委員（2025/01–2026/12）
- International Program Committee Secretary, 8th IFAC Conference on Nonlinear Model Predictive Control
- Publication Co-Chair, 8th IFAC Conference on Nonlinear Model Predictive Control
- Program Committee Member, 13th International Conference on Complex Networks and their Applications
- Associate Editor, 10th Indian Control Conference
- 計測自動制御学会 制御部門 人の理解/誘導で強化される制御システム調査研究会，委員（2024/01–2025/12）
- External Reviewer, Research Grants Council of Hong Kong (2023, 2024, 2026)
- Reviewer, French National Research Agency (2023)
- Program Committee member, 12th International Conference on Complex Networks and their Applications
- Registration Chair, 10th IFAC Symposium on Robust Control Design
- International Advisory Committee Member, 2022 International Conference on Information Technology
- Associate Editor, IET Control Theory & Applications (2022–present)
- Associate Editor, Franklin Open (2022–present)
- Program Committee member, 11th International Conference on Complex Networks and their Applications
- 電子情報通信学会 高信頼制御通信研究会，専門委員（2022–2026）
- 学術変革領域研究（A）領域番号20A403「分子サイバネ」専門委員会，委員
- Program Committee member, 10t International Conference on Complex Networks and their Applications
- Technical Program Comittee Member, 7th IEEE International Conference on Network Softwarization
- 電子情報通信学会 情報ネットワーク研究会，専門委員（2021–present）
- 計測自動制御学会論文集委員会和文論文編集委員会, アソシエイトエディター (2021–2024)
- Associate Editor, Journal of The Franklin Institute（2020–present）
- Program & Steering Committee Member, 52nd ISCIE International Symposium on Stochastic Systems Theory and Its Applications
- システム制御情報学会, 編集委員 (2020, 2021)
- 計測自動制御学会 関西支部 庶務幹事（2019, 2020）
- 計測自動制御学会 制御部門 真なるダイナミクスの追求による次世代システム制御理論調査研究委員会，委員（2019/01–2021/12）
- Local Arrangements Vice Chair, SICE Annual Conference 2018
- 電子情報通信学会 高信頼制御通信研究会，幹事補佐（2018, 2019）
- 計測自動制御学会 制御部門 IoT時代に向けたイベントベースト制御調査研究会，委員（2018/01–2019/12）
- Technical Comittee Member, Technical Committee 1.5. Networked Systems, International Federation of Automatic Control (2015–present).
- Associate Editor, 5th IFAC Workshop on Distributed Estimation and Control in Networked Systems

査読 Advanced Robotics; Annual Reviews in Control; Automatica; Applied Mathematics and Computation; Applied Network Science; Applied Soft Computing; Asian Journal of Control; Chaos, Solitons and Fractals; Computer Communications; Developments in the Built Environment; Discrete Mathematics; Engineering Applications of Artificial Intelligence; Entropy; European Journal of Control European Physical Journal B; Expert Systems With Applications; Foundations of Computational Mathematics; IEEE Access; IEEE/ACM Transactions on Computational Biology and Bioinformatics; IEEE/CAA Journal of Automatica Sinica; IEEE Circuits and Systems Magazine; IEEE Control Systems Letters; IEEE Intelligent Systems; IEEE Internet of Things Journal; IEEE Open Journal of Vehicular Technology; IEEE Signal Processing Letters; IEEE Transactions on Automatic Control; IEEE Transactions on Automation Science and Engineering; IEEE Transactions on Big Data; IEEE Transactions on Circuits and Systems I: Regular Papers; IEEE Transactions on Circuits and Systems II: Express Briefs; IEEE Transactions on Cognitive and Developmental Systems; IEEE Transactions on Computational Social Systems; IEEE Transactions on Control of Network Systems; IEEE Transactions on Cybernetics; IEEE Transactions on Fuzzy Systems; IEEE Transactions on Industrial Electronics; IEEE Transactions on Industrial Informatics; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Neural Networks and Learning Systems; IEEE Transactions on Signal and Information Processing over Networks; IEEE Transactions on Signal Processing; IEEE Transactions on Systems, Man and Cybernetics: Systems; IEEE Transactions on Pattern Analysis and Machine Intelligence; IEEE Transactions on Power Systems; IEEE Transactions on Mobile Computing; IET Control Theory & Applications; International Journal of Systems Science; International Journal of Robust and Nonlinear Control; Internet Technology Letters; Journal of Aerospace Engineering; Journal of Biological Dynamics; Journal of Building Engineering; Journal of Computational Science; e Journal of Systems Science and Complexity; Journal of the Franklin Institute; Mathematics and Computers in Simulation; Mathematics of Control, Signals, and Systems; National Science Review; Neural Networks; Neurocomputing; Nonlinear Analysis: Hybrid Systems; Nonlinear Dynamics; Operations Research and Decisions; Optimal Control, Applications and Methods; Philosophical Transactions of the Royal Society A; Physica A; Physics Letters A; PLOS Computational Biology; Science China Information Sciences; SIAM Journal on Control and Optimization; SICE Journal of Control, Measurement, and System Integration; Stochastics and Dynamics; Systems and Control Letters; Research in Engineering Design; 計測自動制御学会論文集; システム制御情報学会論文誌.